# Invertibility at infinity and the correspondence of invertibility and self-adjointness between the two model operators

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*Editorial scope.* the counted statement and its own proof environment only

*Allowed standard prerequisites (editorial).*

- The two first-order operators on the cylinder and the compact side. The operators D_C^+/- on L^2(S_C^+/-) with domains W^2_D(S_C^+/-; P or 1-P) and the model operators tilde D_C^+/- on L^2(tilde S_C^+) with domains W^2_{tilde D_C}(tilde S_C^+; P or 1-P), together with

the projection P of eq def P.. The pairs of operators the proposition relates; declared as the article's standing notation, no external stub beyond the operator-display stubs.

*Source reference.* Original equation 3.1: sources/sigma-2023-023-spectral-asymmetry-and-index-theory-on-manif/pinned-original-source.tex, source lines 692–692. Navigation pointer invoked at line(s) 1014: the counted text cites that numbered equation of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the equation inside that section, derived by the census v3.1 rule for the section-scoped equation counter); the cited equation is an allowed input and none of it is reproduced inside the span.

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*Source reference.* Original equation 3.14: sources/sigma-2023-023-spectral-asymmetry-and-index-theory-on-manif/pinned-original-source.tex, source lines 890–890. Navigation pointer invoked at line(s) 1037: the counted text cites that numbered

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*Source reference.* Original equation 3.4: sources/sigma-2023-023-spectral-asymmetry-and-index-theory-on-manif/pinned-original-source.tex, source lines 749–749. Navigation pointer invoked at line(s) 1046: the counted text cites that numbered equation of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the equation inside that section, derived by the census v3.1 rule for the section-scoped equation counter); the cited equation is an allowed input and none of it is reproduced inside the span.

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*Source reference.* Original lemma 3.6: sources/sigma-2023-023-spectral-asymmetry-and-index-theory-on-manif/pinned-original-source.tex, source lines 874–874. Navigation pointer invoked at line(s) 1031, 1052: the counted statement is phrased in notation from that statement of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived by the census v3.1 rule for the shared theorem counter); the cited statement is an allowed input and none of it is reproduced inside the span.

*Source reference.* Original lemma 3.5: sources/sigma-2023-023-spectral-asymmetry-and-index-theory-on-manif/pinned-original-source.tex, source lines 852–852. Navigation pointer invoked at line(s) 1031, 1052: the counted statement is phrased in notation from that statement of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived by the census v3.1 rule for the shared theorem counter); the cited statement is an allowed input and none of it is reproduced inside the span.

**Proposition 3.9.**

(*a*) *The operator $D$ is invertible at infinity in the sense of* (3.1) *if and only if there are $b_0 > 0$ and $u \geq a$ such that for all $s \in \Gamma_c^\infty\big(\widetilde{S}_C\big)$ supported in $(u, \infty)$,*

$$\big\|\widetilde{D}_C s\big\|_{L^2(\widetilde{S}_C, \mathrm{vol}_{B_N} \otimes \mathrm{d}y)} \geq b_0 \|s\|_{L^2(\widetilde{S}_C, \mathrm{vol}_{B_N} \otimes \mathrm{d}y)}.$$

(*b*) *The operator* (3.9) *is invertible if and only if the operator* (3.20) *is.*

(*c*) *The operator* (3.10) *is invertible if and only if the operator* (3.21) *is.*

(d) *The operator* $D_C^- D_C^+$ *on* $L^2\big(S_C^+\big)$*, with domain* $W_D^2\big(S_C^+; P\big)$*, is self-adjoint if and only if the operator* $\widetilde{D}_C^- \widetilde{D}_C^+$ *on* $L^2\big(\widetilde{S}_C^+\big)$*, with domain* $W^2_{\widetilde{D}_C}\big(\widetilde{S}_C^+; P\big)$ *is.*

(e) *The operator the operator* $D_C^+ D_C^-$ *on* $L^2\big(S_C^-\big)$*, with domain* $W_D^2\big(S_C^-; 1-P\big)$ *is self-adjoint if and only if the operator* $\widetilde{D}_C^+ \widetilde{D}_C^-$ *on* $L^2\big(\widetilde{S}_C^+\big)$*, with domain* $W^2_{\widetilde{D}_C}\big(\widetilde{S}_C^+; 1-P\big)$ *is.*

*Proof.* Part (a) follows from Lemmas 3.5 and 3.6.

For any $s \in \Gamma^\infty\big(\widetilde{S}_C\big)$, we have

$$(Ts)|_N = \Phi(a) s|_N,$$

because we view $N$ as embedded into $\overline{C}$ as $N \times \{a\}$ and into $N \times [0,\infty)$ as $N \times \{0\}$. Therefore, for any such section, $P(s|_N) = 0$ if and only if $P(Ts|_N) = 0$, and similarly for $1-P$. Hence the operator $T$, together with (3.14), defines unitary isomorphisms

$$\begin{aligned}
&T\colon\ W^1_{\widetilde{D}_C}\big(\widetilde{S}_C^+; P\big) \xrightarrow{\cong} W^1_D\big(S_C^+; P\big),\\
&T\circ\tau\colon\ W^1_{\widetilde{D}_C}\big(\widetilde{S}_C^+; 1-P\big) \xrightarrow{\cong} W^1_D\big(S_C^-; 1-P\big),\\
&T\colon\ W^2_{\widetilde{D}_C}\big(\widetilde{S}_C^+; P\big) \xrightarrow{\cong} W^2_D\big(S_C^+; P\big),\\
&T\circ\tau\colon\ W^2_{\widetilde{D}_C}\big(\widetilde{S}_C^+; 1-P\big) \xrightarrow{\cong} W^2_D\big(S_C^-; 1-P\big).
\end{aligned} \tag{3.23}$$

In the second and fourth lines, we use the fact that by definition (3.4) of $P$, we have

$$P|_{L^2(S^-|_N)} = \tau_+ P^+ \tau_+^{-1},$$

so that the isomorphisms $T \circ \tau$ preserve the given boundary conditions.

Under the isomorphisms (3.23), the pairs of operators in parts (b)–(e) correspond to each other. Here we again used Lemmas 3.5 and 3.6, and also Lemma 3.7. □